

VisTumor: Explainable Cancer Detection in Histopathology Images Using Grad-CAM and Saliency Mapping

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Abstract

Histopathology image analysis is critical for cancer diagnosis. Despite breakthroughs in image classification technology in deep learning models, they have yet to be adapted in medical applications because they often lack transparency in their decision-making processes. To address this limitation, we propose VisTumor, an explainable deep learning pipeline in whole slide histopathology images. Our system integrates several modern CNN architectures with Grad-CAM and saliency mapping techniques to provide visual explanations of model predictions. Using the CAMELYON16 dataset, we preprocess gigapixel WSIs into tissue-focused patches. VisTumor aims to bridge the gap between accuracy and explainability in medical AI and explores the potential of CNNs to be used as clinical decision support tools.

1. Introduction

1.1. Background & Motivation

The work from Liu et al. (2017), Selvaraju et al. (2017), and Tjoa & Guan (2020) provides the conceptual foundation for our work. We are closely following the experiment procedure from Liu et al., which demonstrated the efficacy of using deep, pre-trained networks on histopathology images. Their approach has influenced our data pre-processing and model training process. We are applying Grad-CAM, a technique for generating visual explanations for model decision making, introduced by Selvaraju et al. This research directly supports our goal of creating interpretable diagnostic tools. Lastly, our motivation for research is based largely off of the work done by Tjoa & Guan as they offer a comprehensive investigation of the need for transparent and interpretable medical AI applications.

- Liu et al. (2017) – “Detecting Cancer Metastases on Gigapixel Pathology Images”
- Selvaraju et al. (2017) – “Grad-CAM: Visual Explanations from Deep Networks via Gradient-Based Localization”

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- Tjoa & Guan (2020) – “A Survey on Explainable Artificial Intelligence (XAI): Toward Medical XAI”

1.2. Related Work

Examination of histopathological whole slide images (WSIs) is a cornerstone in cancer diagnosis, however, it requires highly trained experts and is extremely labor-intensive. With the advent of convolutional neural networks (CNNs), we have seen significant progress that showcases how deep learning models can accurately detect metastases in lymph node tissue. However, most of the research done in this field has prioritized accuracy over explainability, which offers little insight into the reasons for a diagnosis. Grad-CAM and saliency maps are popular methods for visualizing key points and can be effective for the evaluation of models, but few works have actually integrated these features.

1.3. Plan

We propose VisTumor, a deep learning pipeline for explainable tumor detection in histopathology images. Our system combines a ResNet-based convolutional neural network trained on the Camelyon16 dataset in conjunction with Grad-CAM pooling to generate heatmaps that highlight regions that contribute to the model’s diagnosis. Our main goal is not only to evaluate classification accuracy, but also to test that our model correctly focuses on annotated tumor regions. Making a model accurate and interpretable is essential for trustworthy decision-making, especially with the gravity of a cancer diagnosis.

1. Pre-processing Data: Because of the resolution of WSIs, we must segment the full slides. Here, we will tune the patch and step size.
2. Choose Model Architecture: Our goal is to train a few models based on existing architectures that other researchers have already tuned. This allows us to ver-

ify which model architectures use the right features to classify.

3. **Hyperparameter Tuning:** Our goal is to use hyperparameters that have been found to already work in existing architectures, but we will verify and experiment if we can achieve better results.
4. **Applying Grad-CAM:** After solidifying and tuning multiple model architectures, we will implement Grad-CAM layers to create heatmaps of model attention.
5. **Testing:** We will evaluate the performance of classifications on our test patches, assessing the ability to generalize. Additionally, we will manually evaluate our saliency maps against professionally annotated slide images.
6. **Compare Performance of Existing Models:** Utilize existing models within the histopathology space and evaluate performance against our model in classification. Additionally, we will qualitatively compare saliency maps generated by our model versus existing models to assess interpretability.

1.4. Expected Challenges

1. **Limited Storage and Compute Resources:** As students, we are reliant on the use of Google Colab, which has some constraints. We have limited storage, which may present challenges regarding the dataset CAME-LYON16, since WSIs often exceed gigapixel resolutions, meaning we must load terabytes worth of data. Additionally, we have limited GPU hours, meaning hyperparameter searching must be carefully constructed to stay within usage limits.
2. **Handling High Resolution WSIs:** Whole slide images are extremely large, which makes them infeasible to input as raw pixels. We must divide each WSI into smaller patches to train our model, but this process limits spatial context.
3. **Performance Trade-Off:** Integrating Grad-CAM or attention for explainability introduces extra computational overhead that may impact accuracy or inference speed. There will be a strong priority on balancing the benefits of visual explanations with the cost of overall performance and resource constraints.
4. **Performance Verification:** Although we can verify model diagnosis accuracy through testing, we cannot validate the accuracy of our heat maps or attention visualizations. Those would require analysis and validation from expert pathologists, which goes beyond the scope of this project/class.

1.5. Expected Outcome

Our project aims to develop a deep learning model that not only accurately detects cancerous regions within WSIs of lymph nodes, but also increments on the interpretability through visual explanations. We expect our model to classify whether an individual image patch contains cancerous tissues. Additionally, by using saliency techniques, we aim to generate heatmaps that display the most influential regions in influencing the model's predictions as a means of transparency. The hope is that such heatmaps are used in aiding in clinical interpretations of WSIs. If our model performance is favorable, we may seek expert feedback from pathologists to validate the relevance of said heatmaps.

The project's pipeline is foundational for other deep learning projects with partially annotated data, such as the WSIs being used. We hope that the performance and interpretability of our proposed model can justify a potential clinical use.

2. Methods

2.1. Data

The dataset being used for this project is CAME-LYON16. This dataset is a benchmark dataset designed for the development and evaluation of deep learning models on cancer detection of metastases in histopathological whole slide images (WSIs) of lymph node tissue. This is a small dataset consisting of 400 WSIs split into 270 training examples and 130 testing examples. Each slide is taken from extremely high-resolution microscope imaging (some over 100,000 x 100,000 pixels). This dataset presents some unique challenges because of its high resolution, but it serves as an excellent test bed for our project, which focuses on accurate and interpretable cancer detection.

2.2. Preprocessing

The preprocessing work we have made can be found [here](#).

2.2.1 Patch Extraction

To focus on more relevant anatomy within our dataset, we began by applying automated segmentation masks on the dataset to locate regions of interest.

The first step to this process was randomly sampling patches of 224 x 224 pixels due to the dimensionality of the input to our model. And although the tissue on the slide covered half of the pixels, the other half was consumed by white or grey pixels. They were obviously not part of the tissue, and passing patches as such into our model would hinder its learning. Thus, when we sample patches, we ensured that none of our patches were over a certain ratio of being grey.

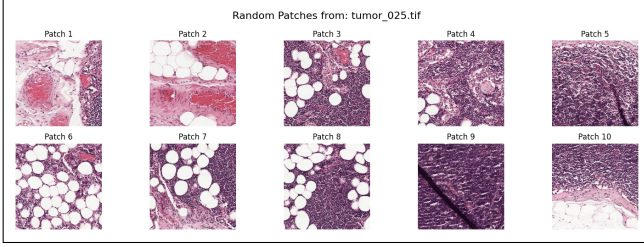


Figure 1. Patches taken during preprocessing without grey patches (non-tissue).

After locating appropriate patches, we will then normalize them. This is to ensure consistent distributions and more efficient convergence during training time. Thus, we will normalize based on the mean and standard deviation of the patches rather than ImageNet’s pre-existing statistics. Each patch was normalized by subtracting the computed mean and dividing by the standard deviation (channel-wise). This allowed us to maintain characteristics of the CAMELYON16 dataset while also ensuring stability.

Please note that at this time, our algorithm to remove non-tissue patches is flawed and needs to be iterated on. As it stands, partially blank patches are present after preprocessing. Additionally because of this, our normalized patches lose many of their color features that may be important to the predictive power of our model. Thus, for the time being, we decided it was best to hold off normalizing the preprocessed data, and saving it for our actual model to do.

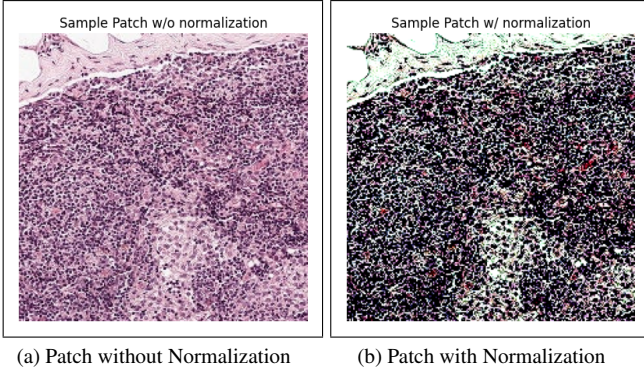


Figure 2. Patch from preprocessed data. One with normalization and one without.

2.2.2 Splitting Data

We used the official training and test splits provided by the Grand Challenge platform. These preset splits ensured comparability with prior works done on the dataset. Samples were not shared between splits, however, 10% of the

training data was segmented off to be used for the validation set. The training set will be made of (To-Be Determined) slides, sampling (To-Be Determined) patches from each.

However, thus far for our preliminary results, we used 5 slides, sampling 100 patches from each.

2.3. Model Architectures

In our project, we will be evaluating the performance and interpretability of 3 modern CNN architectures: ResNet50, DenseNet201, and EfficientNetB7. While these models are robust and widely used in many image classification tasks, we will be leveraging these existing architectures to see if CNNs are strong candidates for future medical imaging applications, in particular.

ResNet introduces residual skip connections that mitigate vanishing gradients and allow it to ignore layers that do not contribute to better results. DenseNet leverages dense connection patterns between layers that allow for better feature reuse and gradient flow through the network. EfficientNet utilizes a compound scaling (depth, width, resolution) method to increase the accuracy and efficiency of CNNs. All of these are viable architectures for our experiment, but we aim to explore whether CNN pipelines are viable for creating explainable models.

Because of our limitations in computing and storage, we will follow the approach used by Liu et. al. in Google’s cancer detection pipeline. Each of our models will be pre-trained on ImageNet, and we will augment and train the last layer on CAMELYON16 to fit our application. The final fully connected layer will be the only layer trained on the dataset, and will be adjusted to produce binary classification outputs as opposed to the 1000 ImageNet classes. We expect these models to perform well enough to assess how well CNN architectures adapt to histopathological cancer detection and how effectively they highlight relevant features. The purpose is not to create new benchmarks or SOTA models.

2.4. Grad-CAM

The method of explainability we have chosen for our pipeline is Grad-CAM. With Grad-CAM, we aim to create localization maps by computing the gradient of the final convolutional feature map. The resulting heatmaps will highlight regions that strongly influence the model’s prediction. As we explore model explainability, this is a useful technique to qualitatively measure model interpretability. While Grad-CAM is great for this experiment, there are several key drawbacks we must take note of. The heat maps may potentially highlight clinically irrelevant regions and may not be able to capture low-resolution or fine-grained features. This is why having strong quantitative results as well as qualitative assessments is important for explanation fidelity.

3. Experiments

3.1. Preliminary Results

For pre-processing the training data, we broke 5 WSIs into 100 patches each, resulting in 500 training patches. 300 patches contain cancerous tissue, and 200 patches contain healthy tissue. For validation, we broke down 2 WSIs into 100 patches each, which resulted in 100 cancerous patches and 100 healthy patches. For our actual experiment, we will train on a greater amount of data and show that our model performs better with scale, but for the sake of testing our pipeline, we have decided to keep our dataset relatively small.

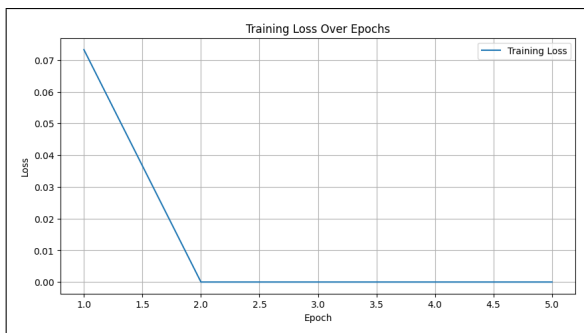
We were unable to train our model because we couldn't figure out how to correctly process our labelled data while training. While we have figured out how to preprocess all of the data, we need to work on correctly integrating it into our model pipelines.

Our evaluation metric for when we fix our data input stream is to use classification accuracy of our model, with cross entropy as our loss function.

[Repo Link](#) and [Data Link](#).

```
Epoch 1, Loss: 0.5859792940314321, Accuracy: 0.974025974025974
Epoch 2, Loss: 0.0, Accuracy: 1.0
Epoch 3, Loss: 0.0, Accuracy: 1.0
Epoch 4, Loss: 0.0, Accuracy: 1.0
Epoch 5, Loss: 0.0, Accuracy: 1.0
```

(a) Training Results over 5 Epochs



(a) Loss Graph

Test Accuracy: 1.0000

(a) Test Results

regards to Learning Rate (λ) and batch size, we decided to go with the default values from PyTorch. These being Learning Rate: $\lambda = 1e-3$ and Batch Size= 32.

- Learning Rate = $1e-3$
- Batch Size = 32
- Epochs = 5
- Optimizer = Adam

Because our primary goal is to test interpretability, we will not be performing random/grid search to find the best hyperparameters and get the best benchmarks. While quantitative model performance is important, our focus is on model interpretability, and not marginal gains based on hyperparameters.

3.2. Hyperparameter Tuning

The hyper-parameters we need to take care of for our model include learning rate for our optimizer, batch size for normalization, epochs for the number of time our model updates its parameters, and the specific optimizer itself. With